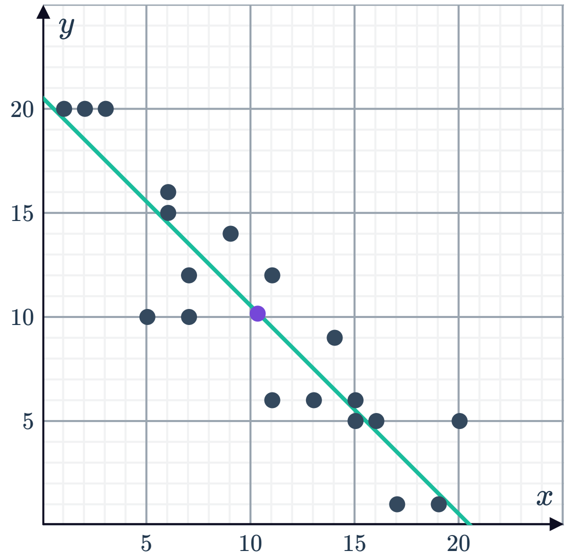
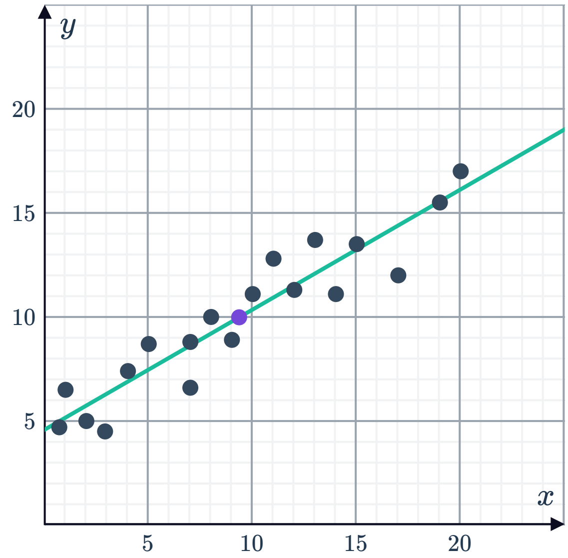
Line of best fit

1

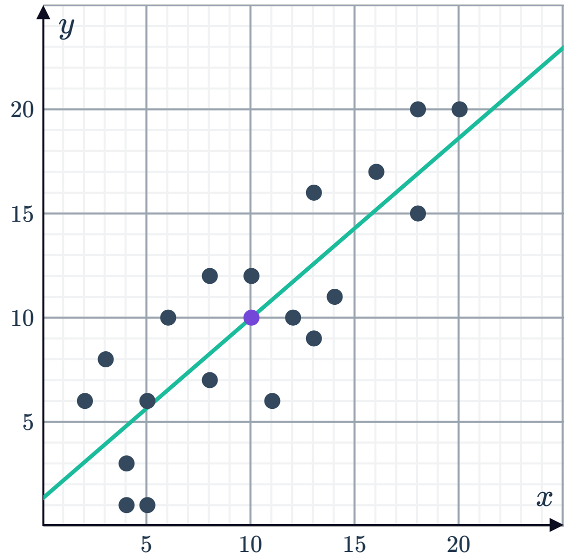
a



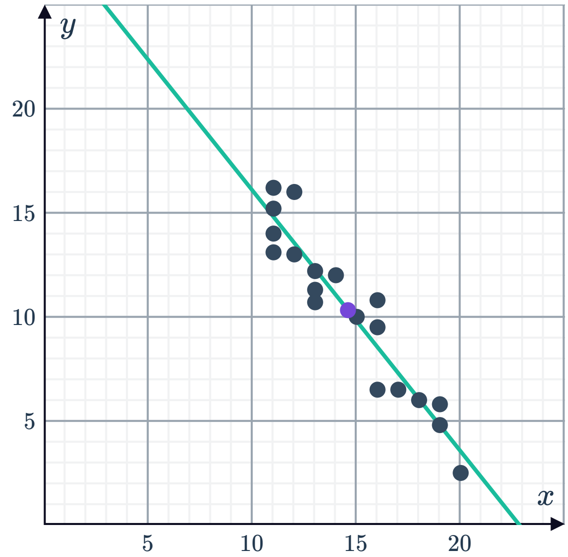
b



c



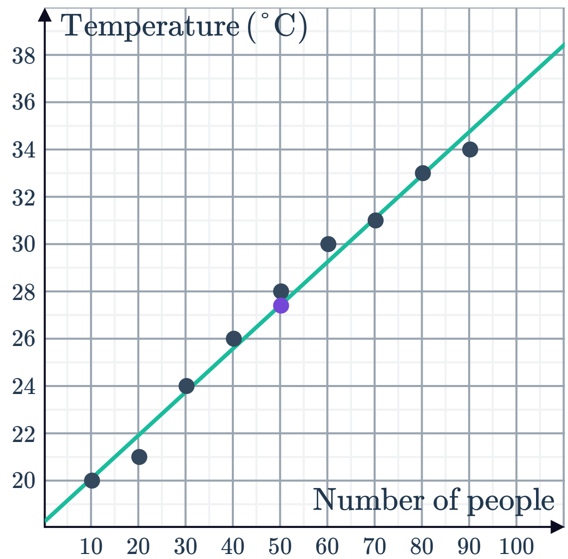
d



2

a $(50, 27.4)$

b

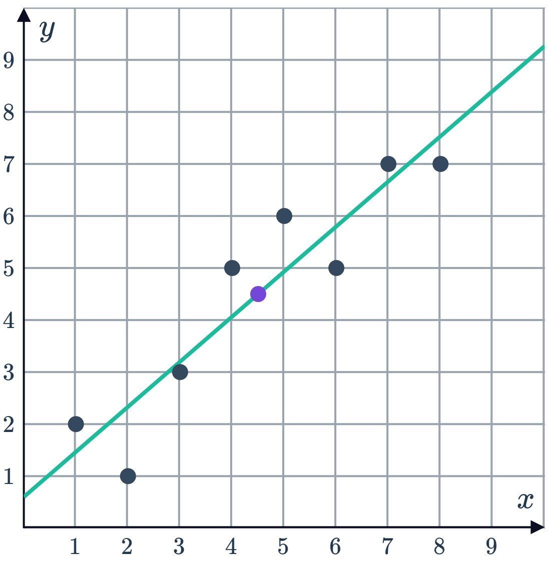


3 $y = 4x - 15$

4 decreasing

5

a



b

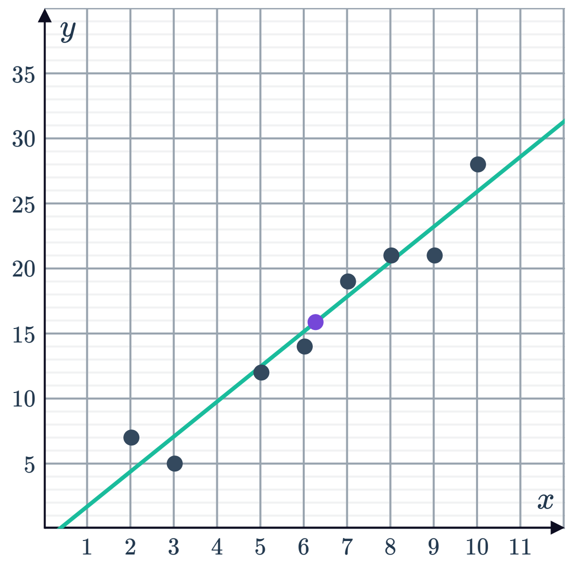
i 4.5

ii Approx. 8.4

6

a Positive

b Example answer:

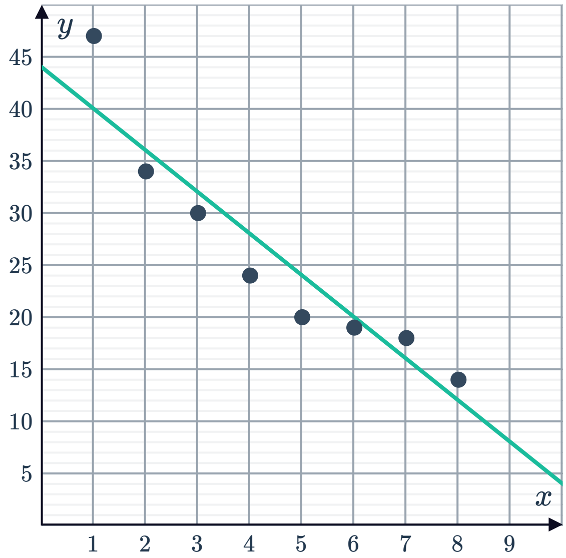


c Example answer: $y = 2.69x - 0.932$

7

a Negative

b Example answer:



c Example answer: $y = -4x + 44$

8

a 3.59

b Positive

c y increases by 3.59 units.

d 6.72

9

a -3.67

b Negative

c y decreases by 3.67 units.

d 8.42

10

a -7.26

b Negative

c T decreases by 7.26 units.

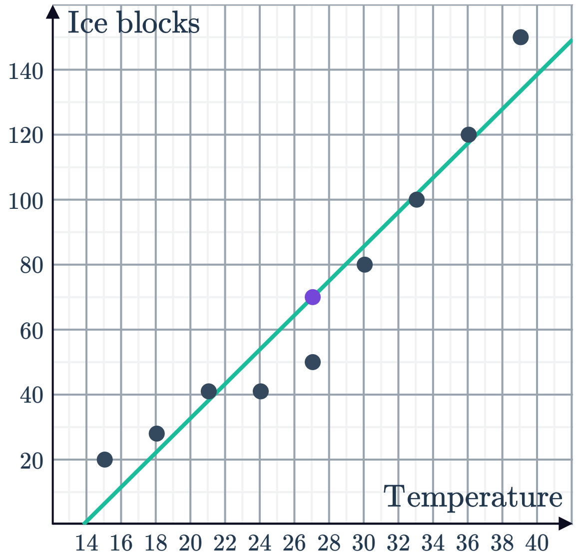
d 4.35

Applications

11 4 knots

12

a

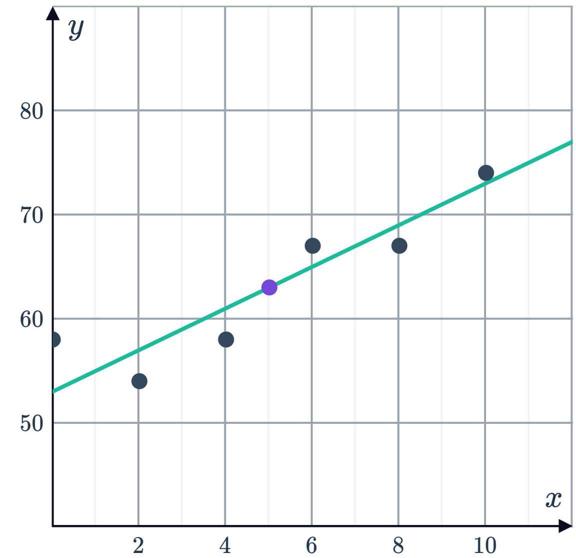


b **i** Approx. 90 **ii** Approx. 150

c Increases

13

a

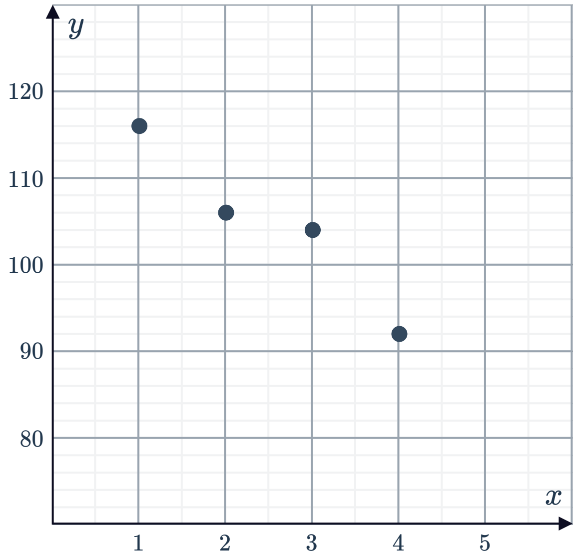


b $(10, 74)$

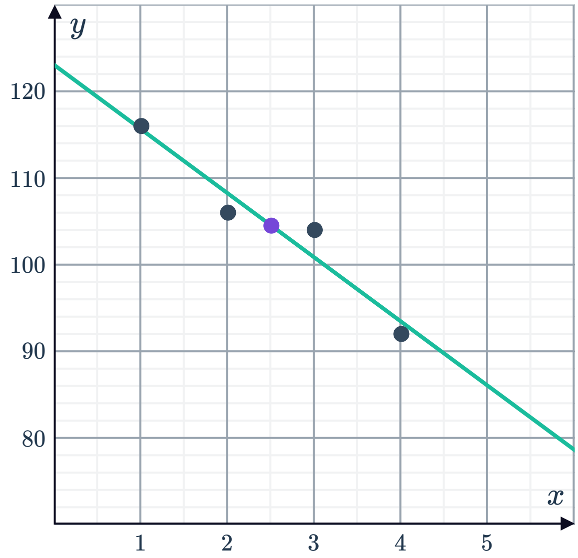
c $(0, 58)$

14

a



b



15

a 27.7 kg

b The y-intercept indicates that when a child is 0 cm , their average weight is 27.7 kg .

c No, a child cannot have a height of 0 cm.

16

a The vertical intercept indicates that when a family has no income, its average spending on dining out is $\$27$.

b No, it does not make sense because the family could not spend $\$27$ dining out each week with an income of $\$0$.

c 0.3

d $\$60$

17

a The vertical intercept indicates that when a student watched no TV, the mark in the exam, on average, was 97%.

b Yes, it is reasonable that a student who spends no time watching TV could get an exam mark of 97%.

c -10

d 35%

18

a 0.09 years per $\$1000$

b 72.55 years

19

a -4

b 3865